

Pricing Variance Swaps

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1. Executive summary

This project estimates the fair variance strike K_{var} of a four-month variance swap using three methods: (i) a Heston analytical approximation, (ii) Monte Carlo simulation under the calibrated Heston model, and (iii) a VIX-style option replication formula. The contract assumptions are $S_0 = 100$, $r = 5\%$, $q = 0$, and $T = 4/12$. The raw option input data consists of 720 call prices across 12 maturities and 60 strikes, with strikes ranging from 70 to 130. The four-month maturity required for the variance swap calculation is available in the dataset.

The calibrated Heston parameters are:

$$v_0 = 0,148066, \kappa = 2,438405, \theta = 0,088444, \sigma = 0,388558, \rho = -0,490976.$$

Please note that in the notation of this report, θ corresponds to the long-run variance parameter

The calibration converged with an RMSE of 0,039564 and a mean error close to zero. The positive Feller margin of 0,280347 suggests that the Feller condition is satisfied.

The resulting fair variance strike estimates are:

Approach	Fair var. strike K_{var}	Vol. strike $\sqrt{K_{\text{var}}}$	Main input
Heston analytical approximation	0,129257	35,95%	Calibrated Heston parameters
Monte Carlo under Heston	0,129877	36,04%	Simulated Heston paths
VIX-style replication	0,122826	35,05%	Raw option data

The analytical Heston and Monte Carlo estimates are close. This is expected because both are based on the same calibrated Heston dynamics. The VIX-style estimate is lower. This is not automatically a numerical failure but rather reflects a different methodology. The VIX-style approach uses the option prices from the dataset directly, while the Heston approaches use a parametric stochastic-volatility model.

2. Data and common assumptions

The variance swap maturity is four months. All rates are assumed to be continuously compounded. The dividend yield is set equal to zero. The annualisation factor for realised variance is 252 trading days per year. Therefore, the number of daily monitoring steps over the four-month life of the variance swap is:

$$n = \frac{4}{12} \times 252 = 84.$$

The realised variance is calculated as:

$$\sigma_R^2 = \frac{252}{84} \sum_{i=1}^{84} \left[\ln \left(\frac{S_i}{S_{i-1}} \right) \right]^2$$

Table 1 — Inputs

Input parameter	Value	Comment
Initial stock price S_0	100	
Risk-free rate r	5,00%	Continuously compounded
Dividend yield q	0,00%	No dividend yield
Variance swap maturity T	1/3 years	4 months
Trading days per year	252	
Daily monitoring steps n	84	$4/12 \times 252$
Option data rows	720	
Number of maturities	12	From 0,0833 to 1 years
Number of strikes	60	Strike grid from 70 to 130
Minimum option price	0,019294	
Maximum option price	35,246221	

The raw option data consists of call prices is used in two different ways. First, the call prices are used to calibrate the Heston model. Second, the call prices are used directly in the VIX-style replication formula.

3. Approach 1: Heston FFT calibration and analytical approximation

3.1 Assumptions and choices

The first approach assumes that the underlying follows the Heston stochastic-volatility model under the risk-neutral measure. The model is parameterised by the initial variance v_0 , mean-reversion speed κ , long-run variance θ , volatility of variance σ , and correlation ρ . The Heston parameter vector is therefore:

$$(v_0, \kappa, \theta, \sigma, \rho)$$

The calibration is performed against the call option prices in the dataset. Call prices under the Heston model are computed using a Fourier-based Carr-Madan style FFT pricing routine. The optimisation objective is based on pricing errors between the call prices from the dataset and model call prices.

The fair variance strike is then approximated using the analytical Heston formula:

$$K_{\text{var}} \approx \frac{1 - e^{-\kappa T}}{\kappa T} (v_0 - \theta) + \theta.$$

This formula reflects the expected average variance over the life of the contract under mean reversion.

3.2 Implementation explanation

Call prices under the Heston model are calculated using the Carr-Madan FFT method. For each maturity, the code evaluates the Heston characteristic function on the Fourier grid used by the Carr-Madan method. The damping parameter α is applied to transform the call payoff into an integrable function. The FFT produces call prices on a log-strike grid. Because the assigned option data is given on a fixed strike grid from 70 to 130, the code then linearly interpolates the resulting FFT call price grid to obtain Heston call option prices at the observed strikes in the dataset. The calibration objective is the RMSE between these interpolated Heston call option prices and the assigned call option prices.

The Heston characteristic function is evaluated under the risk-neutral measure. The Carr-Madan damping parameter is set to $\alpha = 1,5$ and the FFT grid uses $N = 1024$ and $\eta_{\text{FFT}} = 0,25$

The calibration minimises the root mean squared error between call prices in the dataset and Heston model call prices. The optimisation is performed over:

$$(v_0, \kappa, \theta, \sigma, \rho)$$

The parameter bounds used in the calibration are:

Heston Parameter	Lower bound	Upper bound
v_0	10^{-5}	1.0
κ	10^{-3}	20.0
θ	10^{-5}	1.0
σ	10^{-3}	5.0
ρ	-0.999	0.999

Invalid parameter sets are penalised in the objective function. The Feller condition is not used as the calibration objective. It is reported afterwards as a diagnostic check.

The optimisation method is L-BFGS-B¹. The calibration converged after 47 iterations.

Table 2 — Calibrated Heston parameters

Heston Parameter	Description	Calibrated value
v_0	Initial variance	0,148066
κ	Mean-reversion speed	2,438405
θ	Long-run variance	0,088444
σ	Volatility of variance	0,388558
ρ	Stock-variance correlation	-0,490976

The calibrated correlation is negative. This is in line with the theory that lower stock prices are often associated with higher variance. The calibrated initial variance v_0 is higher than the long-run variance θ , so the model implies downward mean reversion in variance over the life of the contract.

¹ Limited memory Broyden–Fletcher–Goldfarb–Shanno algorithm. L-BFGS-B is a gradient-based numerical optimisation method for problems with simple lower and upper bounds on the parameters. It is chosen here because the Heston calibration has explicit parameter bounds and only five parameters, making it more suitable for repeated option-pricing evaluations. Possible alternatives would be differential evolution or least-squares optimisation, but these are generally slower or less directly suited to bounded smooth calibration problems.

Table 3 — Calibration diagnostics

Metric	Value	Interpretation
Optimisation success	Yes	Calibration converged
Optimisation message	Relative reduction convergence	Optimiser stopped normally
Number of iterations	47	Moderate calibration effort
RMSE	0,039564	Main pricing-error measure
MAE	0,028927	Average absolute pricing error
Maximum absolute error	0,157623	Largest individual option error
Mean error	0,000293	Small average bias
Feller margin	0,280347	Positive. Feller condition satisfied

The RMSE is small relative to the option price range in the dataset. The mean error is close to zero, which indicates limited average pricing bias.

The Feller condition is:

$$2\kappa\theta > \sigma^2$$

The diagnostic margin is calculated as:

$$2\kappa\theta - \sigma^2 = 0,280347$$

Since this value is positive, the calibrated parameters satisfy the Feller condition.

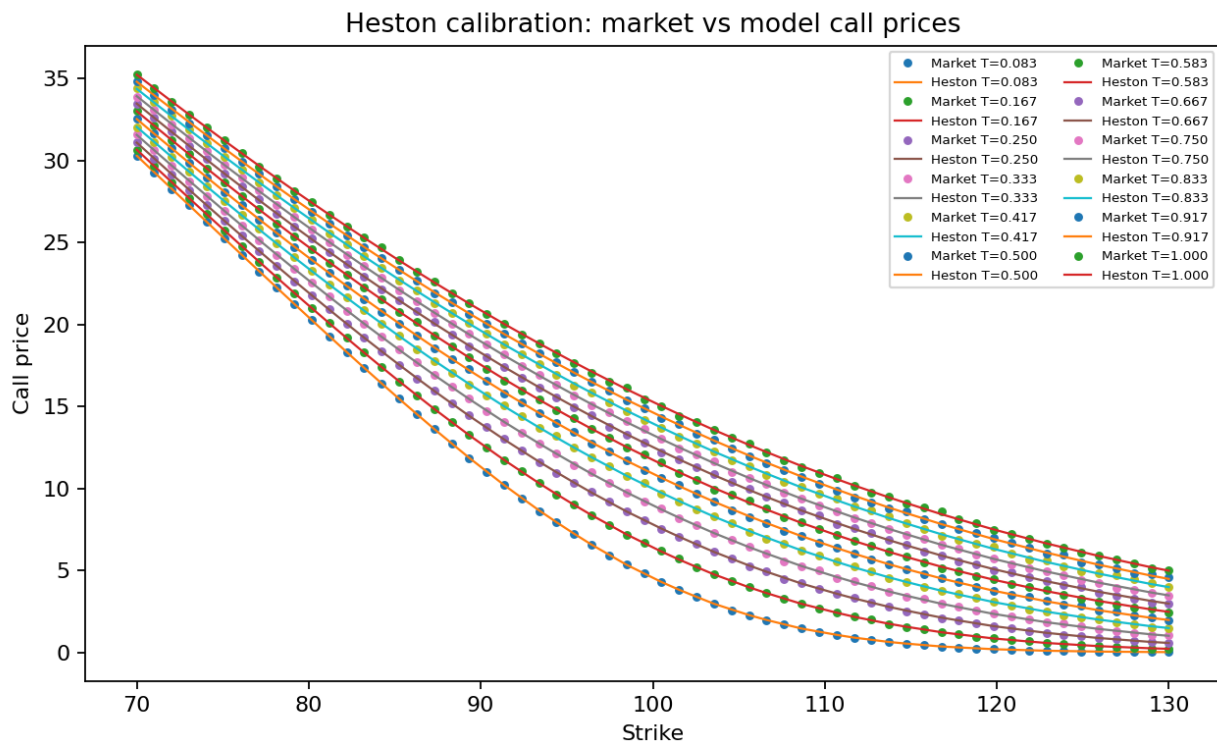


Figure 1 — Heston calibration diagnostic: Assigned call prices versus calibrated Heston model call prices across strikes and maturities. The calibration plot shows that the Heston model prices track the prices from the dataset closely across the strike grid. The fit is not perfect, but the residuals indicate that the remaining errors are limited for the purpose of the assignment.

3.3 Analytical Heston result

Using the calibrated parameters, the mean-reversion weight is:

$$\frac{1 - e^{-\kappa T}}{\kappa T} = 0,684529$$

The fair variance strike from the analytical approximation is:

$$K_{\text{var}} = 0,129257$$

The corresponding volatility strike is:

$$\sqrt{K_{\text{var}}} = 0,359523 = 35,95\%$$

Table 4 — Heston analytical approximation result

Metric	Value
Mean-reversion weight	0,684529
Fair variance strike K_{var}	0,129257
Volatility strike $\sqrt{K_{\text{var}}}$	35,95%

This result is model-dependent. It is interpreted as the fair variance strike implied by the calibrated Heston variance dynamics.

4. Approach 2: Monte Carlo under calibrated Heston

4.1 Assumptions and choices

The second approach uses the calibrated Heston model from Approach 1 and estimates:

$$K_{\text{var}} = \mathbb{E}^Q[\sigma_R^2]$$

This is valid because, under the risk neutral measure, the fair variance strike is the expected realised variance. If the fixed strike is set equal to this expectation, the initial value of the variance swap is 0.

The simulation uses 20.000 Monte Carlo paths and 84 daily time steps, matching the four-month daily monitoring convention used in the assignment. The realised variance is computed path by path using squared log returns and annualised by factor $252/n$. A fixed random seed is used for reproducibility.

4.2 Implementation explanation

The Monte Carlo sampler generates paths for both the stock price and the variance process. At each time step, two standard normal random shocks are generated. These are transformed into correlated shocks so that the correlation between the stock and variance shocks equals the calibrated value: $\rho = -0,490976$.

The variance process is simulated with Euler full truncation. At each step, the code first sets

$$v_t^+ = \max(v_t, 0)$$

The next variance value is then calculated as

$$v_{t+\Delta t} = \max\left(v_t + \kappa(\theta - v_t^+)\Delta t + \sigma\sqrt{v_t^+}\sqrt{\Delta t}Z_{2,t}, 0\right)$$

The stock price is updated with the log-Euler scheme

$$S_{t+\Delta t} = S_t \exp\left[\left(r - q - \frac{1}{2}v_t^+\right)\Delta t + \sqrt{v_t^+}\sqrt{\Delta t}Z_{1,t}\right]$$

The correlated shocks are generated by

$$Z_{2,t} = \rho Z_{1,t} + \sqrt{1 - \rho^2}\varepsilon_t$$

where $Z_{1,t}$ and ε_t are independent standard normal random variables.

The code then calculates daily log returns from consecutive simulated stock prices, squares these returns, sums them over the 84 monitoring dates, and annualises the result using the factor $252/84$.

$$\sigma_R^2 = \frac{252}{84} \sum_{i=1}^{84} \left[\ln\left(\frac{S_i}{S_{i-1}}\right) \right]^2$$

The Monte Carlo estimate of the fair variance strike is the average realised variance across all simulated paths.

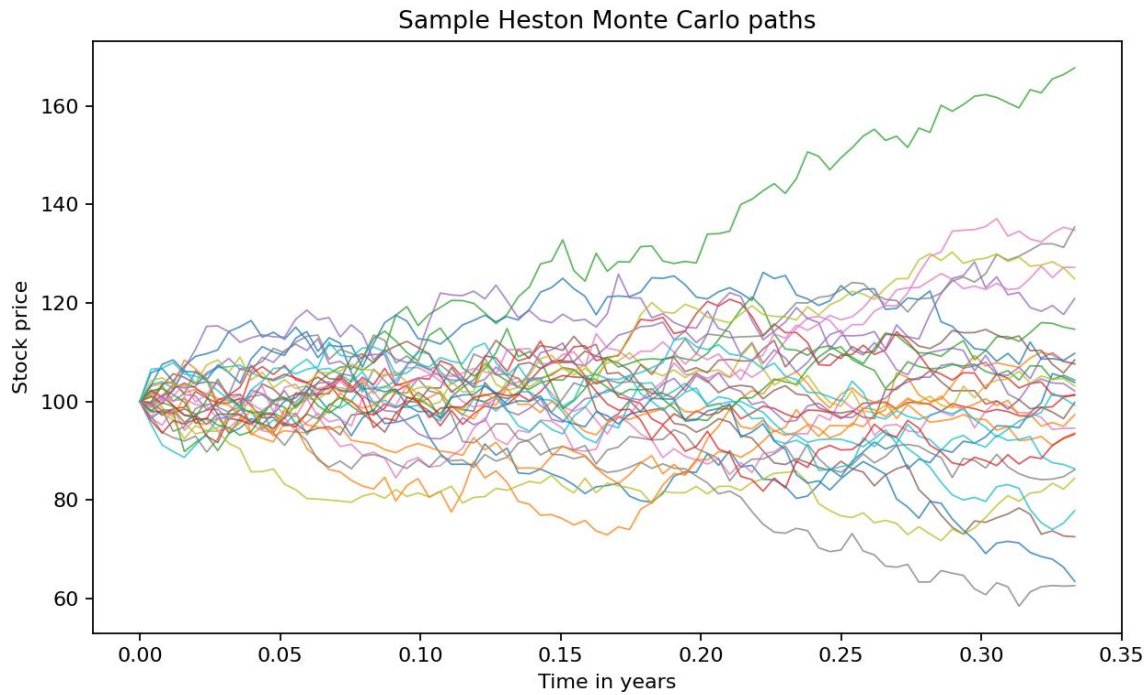


Figure 2 — Sample Heston Monte Carlo paths: Simulated stock price paths under the calibrated Heston model over the four month horizon. This figure documents the path simulation used for the Monte Carlo estimate. The paths start at $S_0 = 100$ and evolve over $T = 1/3$ years.

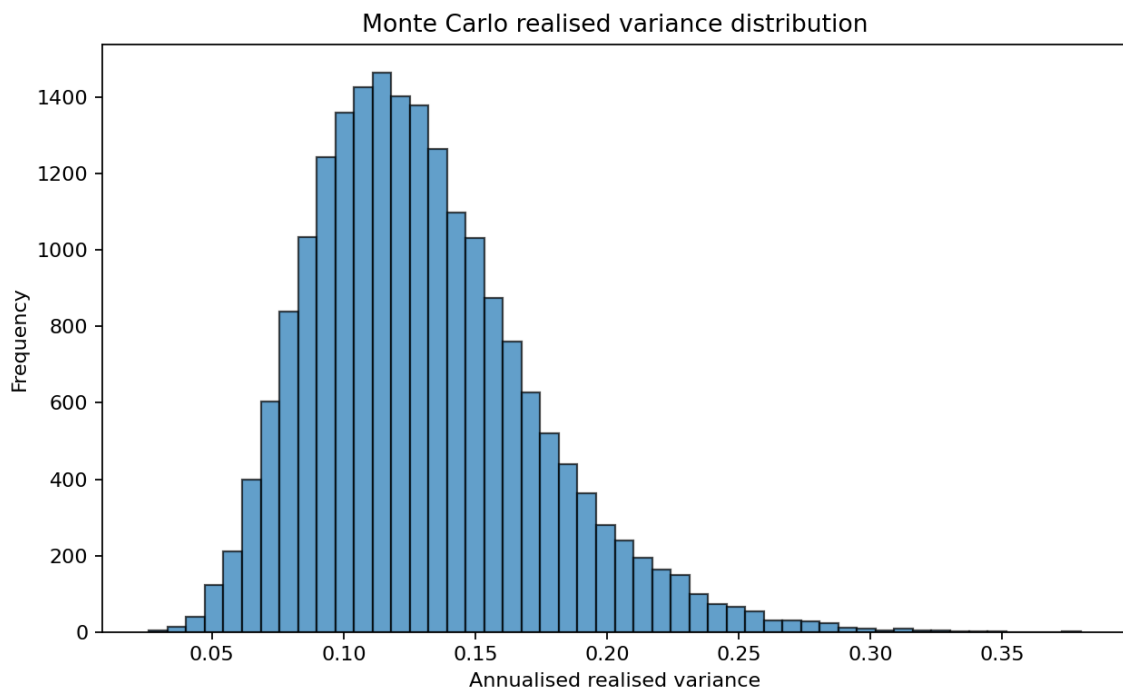


Figure 3 — Monte Carlo realised variance distribution: Distribution of annualised realised variance across 20,000 simulated Heston paths. The realised variance distribution is right-skewed. This is expected because variance is non-negative and stochastic volatility can generate paths with unusually high realised variance.

4.3 Monte Carlo result

The Monte Carlo estimate is:

$$K_{\text{var}} = 0,129877$$

The corresponding volatility strike is:

$$\sqrt{K_{\text{var}}} = 0.360385 = 36,04\%$$

The standard error is 0,000299, giving a 95% confidence interval of:

[0,129291, 0,130464].

Table 5 — Monte Carlo result

Parameter	Value
Number of paths	20.000
Number of time steps	84
Fair variance strike K_{var}	0,129877
Volatility strike $\sqrt{K_{\text{var}}}$	36,04%
Standard error	0,000299
95% confidence interval — lower	0,129291
95% confidence interval — upper	0,130464
Mean terminal stock price	101,542178
Mean terminal variance	0,115345

The Monte Carlo result is close to the analytical Heston approximation. The analytical estimate of 0,129257 lies inside the Monte Carlo confidence interval. This is a useful consistency check because both methods use the same calibrated Heston model.

5. Approach 3: VIX-style replication

5.1 Assumptions and choices

The third approach estimates the fair variance strike using a VIX-style option replication formula. Unlike the first two methods. This approach is less model-dependent than the Heston approaches because it uses the option prices from the dataset directly.

The forward price is calculated as:

$$F_T = e^{(r-q)T} S_0.$$

With $S_0 = 100$, $r = 5\%$, $q = 0$, and $T = 1/3$, the forward price is:

$$F_T = 101,680633$$

The strike K_0 is selected as the largest strike below the forward price. In the calculation, this is:

$$K_0 = 101,525424$$

The option strip used in the VIX-style formula contains 60 options at the target maturity.

5.2 Implementation explanation

The calculation first selects the option observations with maturity $T = 4/12$. The dataset contains call prices. For strikes below the forward price, the VIX-style formula requires put prices. These put prices are obtained from put-call parity:

$$P(K, T) = C(K, T) - S_0 e^{-qT} + K e^{-rT}.$$

Since $q = 0$, this becomes:

$$P(K, T) = C(K, T) - S_0 + K e^{-rT}.$$

For each strike K_i the option input $Q(K_i, T)$ is then selected as:

$$Q(K_i, T) = \begin{cases} P(K_i, T), & K_i < F_T, \\ C(K_i, T), & K_i \geq F_T. \end{cases}$$

The strike interval ΔK_i is calculated from the finite strike grid. For interior strikes, it is half the distance between the neighbouring strikes:

$$\Delta K_i = \frac{K_{i+1} - K_{i-1}}{2}.$$

For the lowest and highest strikes, a one-sided interval is used.

The final estimate is calculated as:

$$K_{\text{var}} \approx \frac{2e^{rT}}{T} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i, T) - \frac{1}{T} \left(\frac{F_T}{K_0} - 1 \right)^2.$$

The second term is the correction term for the difference between the forward price and K_0 .

Table 6 — VIX-style replication inputs and result

Parameter	Value
Forward price F_T	101,680633
K_0	101,525424
Number of options used	60
Fair variance strike K_{var}	0,122826
Volatility strike $\sqrt{K_{\text{var}}}$	35,05%

The VIX-style estimate is lower than the Heston-based estimates. This is explained by the methodology. The VIX-style calculation uses the available finite option grid directly. The theoretical replication result is based on a continuum of strikes, while the dataset only covers strikes from 70 to 130.

6. Comparison of results

6.1 Numerical comparison

Table 7 — Final comparison of variance swap strikes

Method	Fair variance strike K_{var}	Volatility strike $\sqrt{K_{\text{var}}}$
Heston analytical approximation	0,129257	35,95%
Monte Carlo under Heston	0,129877	36,04%
VIX-style replication	0,122826	35,05%

The Heston analytical and Monte Carlo results differ by:

$$0,129877 - 0,129257 = 0,000621$$

This difference is consistent with the Monte Carlo confidence interval. The analytical Heston value lies inside the Monte Carlo 95% interval:

$$[0,129291; 0,130464]$$

The VIX-style estimate is lower than the Monte Carlo estimate by:

$$0,129877 - 0,122826 = 0,007051$$

And lower to the analytical Heston estimate by:

$$0,129257 - 0,122826 = 0,006431$$

In volatility terms, the VIX-style estimate is 35,05%, compared with 35,95% for the analytical Heston approximation and 36,04% for the Monte Carlo estimate.

The dataset contains only call prices. Therefore, for strikes below the forward price, the code constructs synthetic put prices using put-call parity with the same S_0 , r , q , and T as in the assignment.

6.2 Method comparison

Table 8 - Method comparison

Approach	Model dependence	Strengths	Limitations	Interpretation
Heston analytical approximation	High	Fast; directly linked to calibrated variance dynamics	Depends on Heston assumptions; uses an approximation for expected average variance	Heston-implied benchmark fair strike
Monte Carlo under calibrated Heston	High	Directly estimates expected realised variance from simulated paths	Simulation error; depends on discretisation and variance truncation scheme	Numerical check of the Heston fair strike
VIX-style replication	Lower	Uses option prices directly; close to variance replication logic	Sensitive to finite strike range, strike spacing, missing tails, and K_0 choice	Option-strip-implied fair strike

6.3 Assessment of consistency

The three results are coherent.

The strongest internal check is the agreement between the analytical Heston approximation and the Monte Carlo estimate. Both methods use the same calibrated Heston model. The analytical method calculates expected average variance from the Heston variance process, while the Monte Carlo method estimates realised variance path by path. Their closeness indicates that the simulation is consistent with the calibrated model.

The VIX-style estimate is lower. This does not imply that it is wrong. It is a different estimator. It uses the option dataset directly and is affected by the available strike range, strike spacing, and the absence of strike tails outside 70 to 130. By contrast, the Heston approaches impose a smooth parametric model fitted to the option surface.

The calibration diagnostics support the use of the Heston model for the first two approaches. The RMSE is 0,039564, the mean pricing error is close to zero, and the Feller condition is satisfied. However, this does not prove that Heston is the true data-generating model. It only shows that the calibrated Heston model gives a reasonable fit to the option prices in the dataset.

7. Conclusion

This report estimated the fair variance strike K_{var} of a four-month variance swap using three methods.

The first method calibrated a Heston stochastic-volatility model to the dataset using a Carr-Madan FFT call pricer. The calibration converged successfully and produced the following parameter set:

$$v_0 = 0,148066; \kappa = 2,438405; \theta = 0,088444; \sigma = 0,388558; \rho = -0,490976$$

The analytical Heston approximation gave:

$$K_{\text{var}} = 0.129257, \sqrt{K_{\text{var}}} = 35,95\%$$

The second method simulated 20.000 Heston paths with 84 daily time steps and calculated annualised realised variance from daily log returns. The Monte Carlo estimate was:

$$K_{\text{var}} = 0.129877, \sqrt{K_{\text{var}}} = 36,04\%$$

The 95% confidence interval was:

$$[0,129291; 0,130464]$$

The analytical Heston estimate lies inside this interval.

The third method used a VIX-style replication formula on the four-month option strip. The result was:

$$K_{\text{var}} = 0,122826, \sqrt{K_{\text{var}}} = 35,05\%$$

The VIX-style result is lower than the two Heston-based results. The difference is best explained by the distinction between a parametric stochastic-volatility model and a direct finite-grid option replication method.

Overall, the results are consistent. The Heston analytical and Monte Carlo approaches provide mutually supportive model-based estimates. The VIX-style method provides an independent option-strip-based estimate and highlights the effect of using a finite set of strikes.

8. Reproducibility in Python

The calculations were implemented in Python and submitted separately from the PDF report. The main script `run_assignment.py` runs the full workflow: data validation, Heston calibration, analytical Heston variance strike calculation, Monte Carlo simulation, and VIX-style replication.

The project can be rerun with `python run_project.py`. The outputs are written to subfolder `outputs / summary_results.json`. The detailed strike-level contributions are exported in `vix_replication_table.csv`. Monte Carlo uses seed 12345, 20.000 paths, and 84 time steps. The code only makes use of NumPy, SciPy, Pandas, and Matplotlib.

The code is organised into separate modules following the workflow of the report.

1. `config.py` defines the common contract inputs and numerical settings used throughout Sections 2–5.
2. `data.py` loads and validates the assigned option dataset described in Section 2.
3. `heston_fft.py` implements the Heston characteristic function and Carr-Madan FFT call pricer described in Section 3.2.
4. `calibration.py` uses this pricer to calibrate the Heston parameters in Section 3.2.
5. `variance_swap.py` calculates the analytical Heston fair variance strike in Section 3.3.
6. `monte_carlo.py` implements the Heston path simulation used in Section 4.
7. `vix.py` performs the VIX-style replication described in Section 5.
8. `plots.py` generates figures 1 in section 3.2 and figures 2 and 3 in section 4.2
9. `run_project.py` executes the full workflow and exports the final results.